Aluminum Nitride Thin Film Based Reconfigurable Integrated Photonic Devices

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***Abstract*—In the past few decades, silicon photonics with complementary metal-oxide-semiconductor (CMOS) process compatibility has been well developed and successfully applied to commercial products. Due to the demand for high-fidelity and high-speed optical interconnect, high-resolution sensing, and information processing, hybrid integrating more existing materials to improve performance or introduce novel functionalities with post-CMOS technology becomes essential. Aluminum nitride (AlN) is a popular piezoelectric material widely investigated in the field of Micro-Electro-Mechanical System (MEMS). It has been studied and integrated into photonic integrated circuits (PICs) utilizing the mechanics at micro/nano scale in recent years. Here, we review the recent development of AlN thin film properties and processing. The AlN piezoelectric MEMS transducer technologies are then described, including flexural, surface acoustic wave (SAW), Lamb wave resonators (LWR), and bulk acoustic wave (BAW). After that, photonic devices on hybrid Si-AlN and pure AlN thin film platforms are presented. Finally, the outlooks and perspectives of AlN thin film based reconfigurable integrated photonics are provided.**

***Index Terms*—Aluminum nitride, hybrid integration, reconfigurable devices, silicon photonics.**

I. Introduction

SILICON photonics (SiPh), with complementary metal-oxide-semiconductor (CMOS) process compatible materials like silicon (Si), silicon nitride (SiN_x), etc., is one of the most promising platforms for photonic integrated circuits (PICs). A variety of devices for sensing [1], [2], [3], [4], [5], [6], [7], [8], computing [9], [10], [11], [12], [13], [14], and interconnecting [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26] have been demonstrated. Silicon-on-insulator (SOI) based PICs are well studied with versatile functionality due to high refractive index contrast and compact device size. However, high contrast also increases sensitivity to sidewall roughness and leads to a relatively high propagation loss, approximately 1.5-2.5 dB/cm [27]. Multi-mode waveguides [28], [29], [30] may address this issue at the cost of compactness. Moreover, two-photon absorption [31] limits the maximum propagating optical intensity. Compared with Si, SiN_x has a much lower refractive index and exhibits a lower propagation loss of $\sim$ 0.6 dB/cm in single-mode waveguides [32] and 1 dB/m in multi-mode waveguides [33], [34]. In addition, the transparency window of SiN_x can cover from the ultraviolet (UV) wavelength to mid-infrared (MIR).

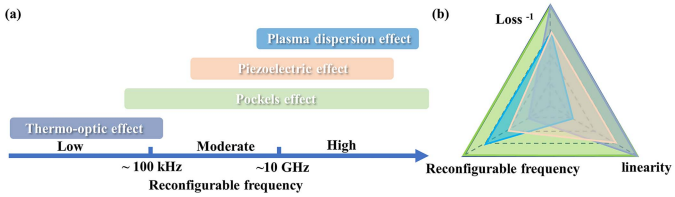


Fig. 1. (a) Modulation speed range of applications based on different tuning mechanisms. (b) Performance of different mechanism-based devices.

The concept of reconfigurable devices has been proposed to increase flexibility and functionality using phase tuning devices. Based on the thermo-optic (TO) effect [35], piezoelectric effect [36], Pockels effect [37], and plasma dispersion effect [38], the TO tuning, acousto-optic (AO) tuning, electro-optic (EO) tuning devices show a broad range of applications in different materials. As shown in Fig. 1, the TO tuning devices have a low reconfigurable frequency, while the EO tuning devices using the Pockels effect or plasma dispersion effect possesses a higher modulation speed. Unlike the Pockels effect, the plasma dispersion effect-based devices have a relatively high loss and poor linearity due to the carrier absorption. The AO tuning devices based on the piezoelectric effect have the advantage of low power consumption.

The commonly used tuning mechanisms in SiPh are the TO effect and plasma dispersion effect. The bandwidth of silicon TO devices can reach approximately 200 kHz, and the insertion loss (IL) is below 0.5 dB. Hence it is widely used as phase

Manuscript received 12 July 2022; revised 13 February 2023; accepted 13 February 2023. Date of publication 15 February 2023; date of current version 11 April 2023. The work was supported in part by the National Natural Science Foundation of China (NSFC) under Grants 61874073 and 61705099, in part by the Natural Science Foundation of Shanghai under Grants 21ZR1443100 and 23ZR1442400, in part by the Science and Technology Commission of Shanghai Municipality under Grant Y7360k1D01, and in part by Lingang Laboratory under Grant LG-QS-202202-05. *(Zhifang Luo and Aoxue Zhang contributed equally to this work.) (Corresponding authors: Tao Wu; Yi Zou.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/JSTQE.2023.3245290.

Digital Object Identifier 10.1109/JSTQE.2023.3245290

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TABLE I
SUMMARY OF MATERIALS' PROPERTIES

Properties	Silicon	Silicon Nitride	Aluminum Nitride	Lithium Niobate
Bandgap	1.1 [49]	N/A	6.1 [50]	4.2 [51]
Transparency window (μm)	1.2-14 [52]	0.47 –6.7 [53]	0.2 - 13.6 [54]	0.35-5 [55]
Refractive index @ 1550 nm	3.47 [52]	1.9-3.1 [56]	n_e=2.16 [57]	n_e=2.13 [58]
			n_o=2.12 [57]	n_o=2.23 [58]
Crystal structure	Diamond	N/A	Wurtzite	Trigonal
Electro-optic coefficient (pm/V)	N/A	N/A	r_{33} = ~1 [59]	r_{33} = 30.8 [60]
			r_{13} = ~1 [59]	r_{13} = 8.6 [60]
Photoelastic coefficient	p_{11} = -0.094 [61]	p_{12} = 0.047[62]	p_{11} = -0.0086 [63]	p_{11} = −0.026 [64]
	p_{12} = 0.017 [61]		p_{12} = -0.03 [63]	p_{12} = 0.088 [64]
	p_{44} = -0.051 [61]		p_{44} = -0.044 [63]	p_{44} = 0.145 [64]
Piezoelectric coefficient (pm/V)	N/A	N/A	d_{33} = 5.53 [65]	d_{33} = -43.70 [58]
			d_{13} = -2.65 [65]	d_{13} = -4.64 [58]
Thermal conductivity ($W \cdot m^{-1}K^{-1}$)	145 [66]	29 [67]	285 [65]	5.6 [68]
Thermal-optic coefficient ($\times 10^{-5}$/K)	18 [69]	1.7 [70]	2.32 [65]	3.2 [71]
Thermal expansion coefficient ($\times 10^{-6}$/K)	2.6 [72]	3.3 [67]	α_a = 4.2 [73]	α_a = 14.1 [60]
			α_c = 5.3 [73]	α_c = 4.1 [60]
CMOS process compatibility	Yes	Yes	Yes	No

shifters in applications that do not have strict speed requirements. Plasma dispersion effect-based devices can achieve a modulation efficiency with $V_\pi \cdot L$ below 0.3 V·cm [39], [40], [41]. With a well-designed doping profile, structure, and circuit, the bandwidth could be up to 110 GHz [42], [43]. A system with a data transmission rate exceeding 800 Gbps was reported in 2020 [44]. Although the plasma dispersion effect is efficient, the free carrier induces excessive loss, and the modulation is highly nonlinear. Hence, complex modulation schemes with multiple silicon modulators require pre-compensation [45]. Besides that, the excessive loss and carrier frozen phenomenon limit its application in quantum computing which generally requires quantum state preservation with little perturbation. The piezoelectric effect and Pockels effect are unavailable in the SOI platform due to the centrosymmetric diamond crystal structure of single-crystal silicon. Although Jacobsen et al. [46] have demonstrated using a strained SiN_x layer to break the Centro symmetry of Si, which makes the Pockels effect available on the silicon platform, the Pockels effect is still weak and requires high voltage bias as well as long path length. Without introducing new materials, the SiN_x platform is only suitable for building passive devices due to lacking a tuning mechanism. Even TO tuning, the low TO coefficient makes SiN_x TO devices low efficiency [47], [48]. New material platforms with different tuning approaches are highly demanded to pursue high-performance reconfigurable devices.

In the past decade, lithium niobate ($LiNbO_3$) and aluminum nitride (AlN) are emerging materials for integrated photonics research for their strong piezoelectric effect, which provides additional tuning approaches and can be hybrid integrated with MEMS-based devices. Furthermore, both materials possess a strong Pockels effect, essential for high-speed modulation with good linearity. We summarize the materials' properties in Table I [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73]. A variety of photonic devices based on $LiNbO_3$ bulk or thin-film platforms have been demonstrated due to their superior optical and piezoelectric properties. By virtue of its strong Pockels effect, a modulator in $LiNbO_3$ thin film platform with CMOS-compatible drive voltage was demonstrated by Wang et al. in 2018 [74]. Driven by 200 mV peak-to-peak voltage, the transmission data rate achieves 70 Gbps with approximately 0.37 fJ/bit energy consumption. $LiNbO_3$ thin film modulators are suitable for complicated modulation schemes due to their good linearity. A 16-quadrature amplitude modulation (QAM) up to 240 Gbps was demonstrated by consisting of two $LiNbO_3$ thin film modulators [75]. However, the $LiNbO_3$ thin film process is incompatible with CMOS or current silicon photonics. Further heterostructure integration or hybrid packaging is required for future high-performance PICs. Several Si- $LiNbO_3$ hybrid devices have been reported [76], [77], [78], [79], [80]. Considering the fabrication complexity and cost, the CMOS process compatible AlN thin film has become one of the most promising candidates for hybrid platform integration.

The wurtzite crystal structure of AlN thin film enables piezoelectric and nonlinear optical properties. The rational piezoelectric coupling coefficients and low dielectric loss make AlN film a solid candidate in the MEMS field. Many different types of MEMS transducers have been reported on AlN thin film, such as flexural mode transducers [81], [82], surface acoustic wave (SAW) resonators [83], Lamb wave resonators (LWRs) [84], and bulk acoustic wave (BAW) resonators [85], etc. Several hybrid devices, such as AO phase tuning devices [86], [87], [88], [89], [90], [91], [92] and EO phase tuning devices [59], [93], [94], have been investigated. The bandwidth of these phase tuning devices ranges from direct current to tens of GHz. Moreover, AlN thin films exhibit some unique optical properties. With a large bandgap (6.1eV) [95], the transmission spectrum covers UV, visible (VIS), near-infrared (NIR), and MIR regimes. In addition, the refractive index of AlN is approximately 2.14 at 1550 nm [65], and a low loss waveguide with 0.42 dB/cm can be achieved [94]. Based on the properties mentioned above, the AlN thin film can not only act as a MEMS layer hybrid integrating with the SiPh platform but also form a monolithic platform for multi-physics devices.

This article gives an overview of the AlN-based reconfigurable integrated photonics devices. We briefly discuss the challenges faced in fabricating AlN-based functional devices in Section II. Different deposition and etching methods are shown in this section. In Section III, we introduce the typical AlN MEMS

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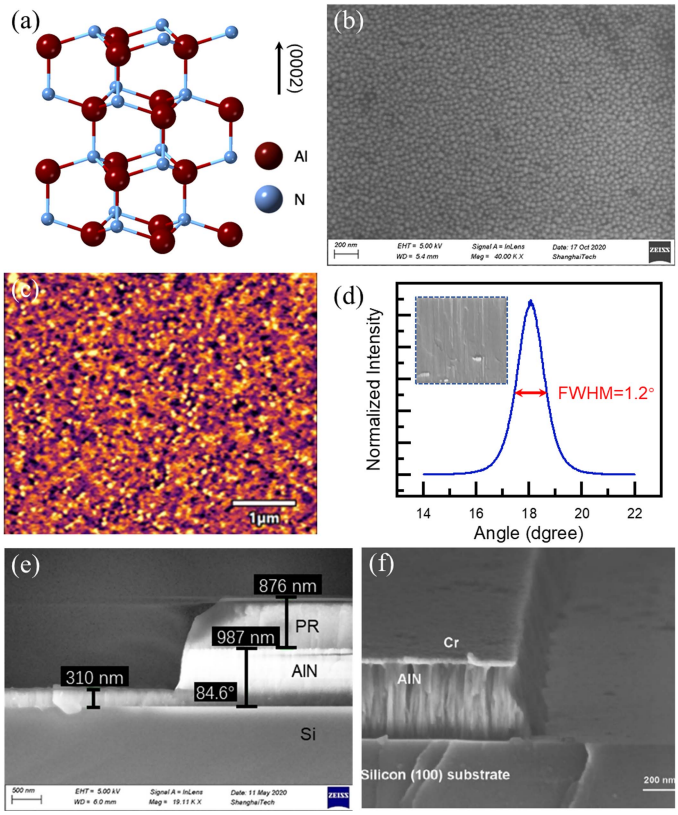


Fig. 2. (a) Crystal structure of N-polar AlN. (b) SEM top view of AlN (0002-polar) surface, and (c) AFM roughness image of AlN, the RMS value is 970 pm. (d) XRD rocking curve of AlN thin film with an FWHM of 1.2°, the inset is the cross-section view of (0002) AlN. (b)–(c) reproduced with permission [96] © 2021, Elsevier. The anisotropically etched image of AlN thin film utilizing (e) dry etching, reproduced with permission [96] © 2021, Elsevier, and (f) wet etching technology, reproduced permission from [97] © 2005, Elsevier.

devices, such as flexural resonators, SAW resonators, BAW resonators, and LWRs. Then, several piezoelectrically coupled optical reconfigurable devices, proposed in the SiPh platform and Si-AlN hybrid platform, are overviewed in Section IV. The AlN thin-film based photonic devices and MEMS reconfigurable photonic devices are introduced in Section V. Lastly, a summary and perspective on some well-noted topics of AlN-based hybrid devices are presented.

II. AlN Thin Film Processing

AlN film is constructed in the wurtzite structure, including the alternating layers of Al and N atoms. Thus, the surface plane of AlN can be either N-polar or Al-polar (Metal-polar). Fig. 2(a) shows the crystal structure of N-polar AlN [96]. The piezoelectricity of AlN is created by non-central symmetry in such a wurtzite structure. The generated charges can be observed when a force is applied to the AlN film (direct piezoelectricity), while the strain can be detected when applying AC electrical fields (inverse piezoelectricity). Deposition and etching of AlN thin films are two challenging processes in fabricating optical or acoustic devices within micro/nano scale. The quality of AlN thin film is generally characterized by surface morphology and crystalline rocking-curve [96] (Fig. 2(b)–(d)), while AlN film sidewall profile and smoothness would affect the acoustic and optical performance [97] (Fig. 2(e)–(f)). The recent technologies to face such challenges are discussed below.

A. AlN Thin Film Depositions

The pursuit of high-performance integrated optical and acoustic properties drives the AlN thin film deposition. Different deposition technologies, including reactive magnetron sputtering, metalorganic vapor phase epitaxy (MOVPE), metal-organic chemical vapor deposition (MOCVD), and molecular beam epitaxy (MBE), have been developed to obtain AlN thin films with high crystal quality, smooth surface morphology, well stress control, and high piezoelectric coefficients. The MBE and MOVPE technologies are usually carried out at high temperatures (800∼900 °C), which have been widely used in the application of photonic devices due to their superior film quality and smooth epitaxial surface [98]. However, deposition at such high temperatures faces a challenge in the hybrid system integration. MOCVD technology requires an even higher temperature of the substrates (about 1180 °C) [99], restricting the development of variable piezoelectric MEMS transducers. In comparison, reactive magnetron sputtering technology is ideal for the multi-system integrated process with sputtering temperature below 450 °C [100]. Table II summarizes the AlN film qualities and related applications of several common deposition technologies and is listed below [101], [102], [103], [104], [105], [106], [107], [108], [109]. The full width at half maximum (FWHM) of the rocking curve obtained by X-ray diffraction (XRD) technology is frequently utilized to characterize the crystalline quality of films. Generally, the nonlinear/EO/AO properties of thin films are strongly relied on the c-axis (0002) crystal orientation [110]. The decrease in FWHM indicates better crystalline quality. Single crystal can be obtained by utilizing epitaxial technologies, while the magnetron sputtering always deposit quasi-single crystal. Although the single crystalline AlN thin film is preferred for its strong polarization (FWHM$<1°$), the challenge in industrial fabrication limits its further applications. Thus, AlN with a quasi-single crystalline structure is an alternative with its moderate polarization and less manufacturing difficulty.

B. AlN Thin Film Etching

Patterning of the AlN thin film is crucial in acoustic and optical applications. The demand for micro/nano devices with high Q-factor pushes forward the different wet and dry etching technologies of AlN thin films. The smooth surface and steep angle after the etching process are essential to achieving low loss optical devices and high electromechanical-coupling coefficient (k_t^2) for acoustic devices [111].

Although AlN has excellent chemical stability, some acid or base solutions are able to etch it. N-/Metal- polar are the two types of stacks in the wurtzite structure of AlN. Kirste et al. show that the etching rate is polarity-dependent [112]. At room temperature, the reaction between acids and AlN is minimal. However, as the temperature goes high, the reaction increases dramatically. Several wet etchants can etch AlN films. The phosphoric acid solution (H_3PO_4) etches AlN film isotropically [113]. One of the most popular base solutions for

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TABLE II
SUMMARY OF ALN DEPOSITION TECHNOLOGIES

AlN deposition technologies	FWHM (°)	Stress (MPa)	Thickness (nm)	Applications [a)]	Mechanical Q	Optical Q
Reactive magnetron sputtering [101]	3.3	N/A	2400	SAW	1970	N/A
Reactive magnetron sputtering [102]	1.57	< 200	160	BAW	930	N/A
Reactive magnetron sputtering [103]	1.9	-200	1000	LWR	1605	N/A
Reactive magnetron sputtering [104]	1.78	-3.8	350	MRR	N/A	1.71×10^5
Reactive magnetron sputtering [105]	2	N/A	330	MRR	N/A	4.4×10^5
Reactive magnetron sputtering [106]	N/A	400	600	MRR	N/A	10^5
MOPVE [107]	~0.75	N/A	1200	MRR	N/A	2×10^6
MOCVD [108]	N/A	N/A	500	MRR	N/A	2.1×10^5
MBE [109]	N/A	N/A	550	MRR	N/A	7.5×10^4

a) SAW, BAW, and LWR are for acoustic applications; MRR (micro-ring resonator) is for optical applications.

TABLE III
SUMMARY OF ALN ETCHING TECHNOLOGIES

Ref.	Mask	ICP power (W)	RF power/voltage	Pressure	Etching gas	Selectivity	Etching rate (nm/min)	Angle (°)
[117]	Ni	500	-310 V	2 mTorr	$Cl_2/BCl_3/Ar$	~30	176	80.1
[118]	SiO_2	700	80 W	0.5 Pa	$Cl_2/BCl_3/Ar$	0.6	77.5	83
[96]	Photoresist	550	80 W	0.5 Pa	$Cl_2/BCl_3/N_2$	0.77	230	84.6
[107]	SiN_x	500	-200 V	0.3 Pa	$Cl_2/BCl_3/Ar$	3	211	80
[119]	SiN_x	500	100 W	0.5 Pa	$Cl_2/BCl_3/Ar$	2.6	~400	N/A

AlN wet etch is potassium hydroxide (KOH), which also shows excellent etching selectivity between N-/Metal- polar AlN thin films. Over 1000 nm/min etching rate under N-polar surface and nearly 2 nm/min under Metal-polar AlN film are observed [114]. Moreover, the etching of the (1100) plane of AlN is more inert than the (0002) plane in KOH. Methylammonium hydroxide (TMAH) is another alternative to KOH, and, same as the KOH, it etches AlN film anisotropically. TMAH (25 wt.%) shows a much higher etching rate of approximately 8000 nm/min than other solutions at a similar temperature (∼ 80 °C) [115].

Dry etching methods of AlN include ion beam etching (IBE), reactive ion etching (RIE), and inductively coupled plasma (ICP) etching. IBE is a pure physical etching method utilizing Ar gas to achieve a reasonable etching rate and profile [116]. A mixture of $Cl_2/BCl_3/Ar$ has been commonly used in RIE and ICP etching. The lack of a biased power source for plasma generation in RIE results in a lower etching rate. Due to the high etching rate, reasonable selectivity, and nearly vertical etching profile, the Cl_2-based ICP etching, combining the large plasma density and energy, is considered the most effective approach for AlN thin film patterning. The recent reported studies of AlN (0002-polar) ICP etching is listed in Table III [96], [107], [117], [118], [119].

III. ALN MEMS BASED DEVICES

In the past decades, due to the demand for low cost, high performance, and small size in different frequency applications, piezoelectric MEMS transducers are the promising candidates for energy harvesters, acoustic transducers, and the next generation of compact mobile devices. Thanks to the moderate electromechanical energy transduction efficiency, high phase velocity, and CMOS-compatible sputtering process, AlN thin films have been widely used in fabricating integrated devices for applications from sensing (inertia, mass, acoustic [120], [121], pressure, and chemical sensors) [122], [123], [124], actuation (switches, speakers) [125], [126], to Radio Frequency (RF) communications (resonators, filters) [127], [128].

If we consider the mechanical mechanisms, the AlN MEMS based devices can be divided based on the structural vibration modes. The device prototype images [81], [82], [129], [130], [131], [132] and displacement fields of four commonly used vibration mechanisms are shown in Fig. 3(a)–(d) utilizing 2D finite element simulation by COMSOL Multiphysics, respectively. The vibration displacement with the fundamental mode of a flexural transducer is shown in Fig. 3(a). Such flexural mode transducers achieve large out-of-plane displacement and typically consist of bi-layer composite layers. The AlN-based SAW transducers typically consist of a piezoelectric layer deposited on a high acoustic velocity or reflective substrate. And the SAW is excited by the interdigitated transducers (IDT) on top of AlN film. The acoustic wave travels along the surface of the piezoelectric layer, as illustrated in Fig. 3(b). Compared to SAW, the acoustic wave that travels along with the in-plane thin film or composite film structure is called the Lamb wave. The Lamb wave piezoelectric transducers are usually formed by the piezoelectric layer sandwiched by the top and bottom electrodes, in which either side can be IDTs or floating metal. The electric field is generated in the piezoelectric layer along the thickness direction when the input RF signal is applied. Then, the Lamb wave is excited and travels along the in-plane direction of the piezoelectric layer, utilizing d_{31} and d_{33} piezoelectric coefficients depending on the IDT geometry and AlN film thickness. Fig. 3(c) shows the displacement field of the LWR fundamental symmetric (S_0) mode. The piezoelectric transducers utilizing the acoustic wave propagating in the solid body are named BAW transducers. The displacement field of a typical film bulk acoustic wave resonator (FBAR) is presented in Fig. 3(d). The general working frequency range of AlN-based transducers is summarized and shown in Fig. 3(e).

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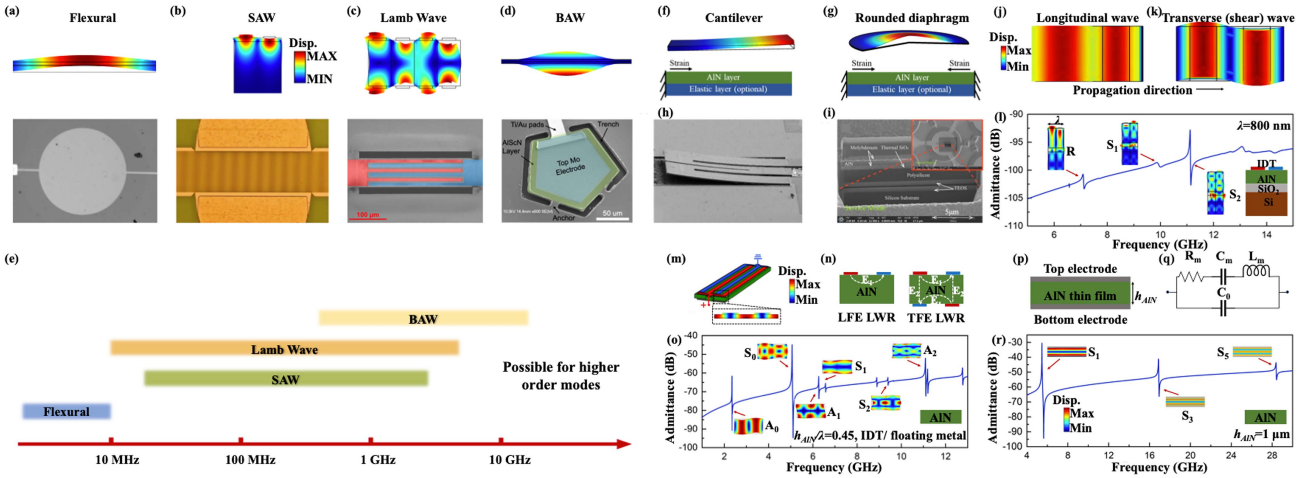


Fig. 3. Displacement fields and images of AlN-based devices utilizing (a) flexural mode, reproduced with permission. [129] © 2021, IEEE. (b) surface acoustic wave, reproduced with permission. [130] © 2021, IEEE. (c) lamb wave, reproduced with permission. [131] © 1969, IEEE. and (d) bulk acoustic wave, reproduced with permission. [132]. © 2020, IEEE. (e) The frequency range of different kinds of AlN-based acoustic devices. The simulated displacement and illustration of (f) cantilever and (g) rounded diaphragm. SEM images of fabricated (h) AlN switch, reproduced with permission. [81] © 2017, IEEE. and (i) the PMUT array, operating at around 5.94 MHz to 8.38 MHz, reproduced with permission. [82] © 2021, IEEE. The displacement of a rectangular grid of (j) longitudinal wave and (k) transverse (shear) wave along the propagation direction. (l) Simulated frequency response of a SAW resonator ($\lambda = 800$ nm, $h_{AlN} = 1$ μm) using FEA simulation. (m) Schematic of the 1-port LWR. Inset shows the FEA of the device displacement fields. (n) Representation of LFE and TFE LWRs, respectively. (o) Spectrum of resonance response of an LFE LWR, as well as the mode shapes. The S_i and A_i represent the symmetric and anti-symmetric modes, respectively. The thickness of the top/bottom electrodes is ignored. (p) Simplified stack of BAW resonators. (q) Equivalent circuit model of BAW resonators. (r) The frequency responses of BAW resonators in the range of 4 GHz to 30 GHz using $h_{AlN} = 1$ μm. The thickness of the top/bottom electrodes is ignored.

A. *Flexural Resonators*

AlN flexural mode resonators have been widely demonstrated in energy harvesters, piezoelectric MEMS microphones or speakers, as well as piezoelectric micromachined ultrasonic transducers (PMUT). The suspended structures could be in the form of a cantilever, clamped-clamped beam, tuning fork, or rounded/squared diaphragm, depending on the operating frequency regime [129], [133], [134]. Finite element analysis (FEA) simulation and mechanics of bilayer laminates of typical simplified one-dimensional flexural resonators (one is the AlN piezoelectric layer, and the other is the elastic layer) are shown in Fig. 3(f)–(i). When an electric field is applied thru the thickness direction, the AlN layer is subjected to a longitudinal strain. Then, the cantilever or diaphragm bends to deflect. As shown in Fig. 3(f)–(i), the maximum displacement occurs at the free end of the cantilever or the center of the diaphragm at the fundamental resonance, while the maximum stress or strain presents near the anchor region, which is usually called the nodal point. Fig. 3(h) and (i) show a typical AlN-based electrostatic switch and AlN PMUT arrays operating in the flexural mode [81], [82]. High-order resonance modes can be excited with careful design of electrode configuration and resonant structures [135]. Optical devices can be co-designed near the position of either maximum displacement or maximum stress, depending on the coupling mechanism.

B. *SAW Resonators*

The SAW was reported by Lord Rayleigh in 1885 for the first time [136], which consists of two basic types of waves, i.e., longitudinal wave and transverse (shear) wave. Both waves propagate along the plane surface of materials with amplitude decaying away from the plane surface. The displacement of the longitudinal wave is parallel to the propagation direction, and the transverse (shear) wave has displacement in the direction normal to the propagation direction and parallel to the wavefront. These two SAWs are illustrated in Fig. 3(j)–(k) utilizing FEA simulation. Rayleigh wave (R) is a surface wave with both longitudinal and surface shear wave components, which is non-dispersive. Sezawa wave (S) is a Rayleigh-type SAW that occurs in a layered structure, which is dispersive and has a series of modes (i = 1, 2, 3, …) with different velocities. The multi-resonance behaviors of the SAW resonator using IDT/AlN/SiO_2/Si configuration are shown in Fig. 3(l). The AC signal is applied to the IDTs to generate SAW, and the SiO_2 layer serves as an acoustic reflector. Under a periodic wavelength (λ) of 800 nm, the resonance frequency of the Rayleigh wave is approximately 7 GHz. As illustrated in Fig. 3(l), a higher S_2 mode resonance is more dominant at approximately 11.1 GHz in this configuration.

C. *Lamb Wave Resonators*

Lamb wave typically combines the bulk shear and longitudinal waves. When a piezoelectric plate resonator with a nanoscale thickness vibrates, the excited acoustic wave may propagate along the plate plane. The schematic of the LWR and its simulated displacement at resonance are shown in Fig. 3(m). The resonating structure can be seen as a suspended rectangular plate with air boundaries on both sides. Two main configurations can be utilized to excite Lamb waves in AlN thin films, lateral

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field excitation (LFE) and thickness field excitation (TFE). As illustrated in Fig. 3(n), as no floating metal exists, the E_1 is excited between top IDTs. For TFE LWR, both E_1 and E_2 are excited, where the E_2 is formed between top and bottom IDTs. The multiple-frequency spectrum of an LFE LWR is presented in Fig. 3(o). Several types of symmetric mode (S_n) and asymmetric mode (A_n) Lamb waves can be exited, depending on the dispersive characteristics of different modes of Lamb waves and the electric-field-induced piezoelectric strain configuration [137]. Therefore, careful consideration should be taken into account towards hybrid integration.

D. BAW Resonators

BAW resonators manufactured by MEMS technology and piezoelectric thin film have attracted much attention for high-frequency applications. FBARs [132] and solidly mounted resonators (SMRs) [138] are two major types of BAW resonators that are well studied in RF oscillators and filter applications. The AlN FBAR is composed of an AlN piezoelectric layer sandwiched by top/bottom electrodes. SMRs consist of acoustic reflectors (Bragg mirrors) under the bottom electrodes. A simplified structure of AlN FBAR is shown in Fig. 3(p). When a sinusoidal voltage with the resonance frequency f_s of the BAW mode is applied to the two electrodes, the dispersive bulk transverse (shear) wave is generated. Symmetric odd modes (S_i, i $= 1, 3, 5, \ldots$) are usually effectively excited in BAW resonators, where the even modes are vanished due to the counteraction of BAWs. As shown in Fig. 3(q), the equivalent circuit of BAW resonators can be simplified using Butterworth-Van Dyke (BVD) model [139], where the C_m, R_m, L_m, and C_0 are motional capacitance, resistance, inductance, and static capacitance, respectively. Fig. 3(r) shows the frequency spectrum of a BAW resonator on the 1-μm-thick AlN thin film. Similar to other resonator techniques, high order resonances, e.g., S_1, S_3, and S_5 modes, exhibit higher frequencies of approximately 5.6 GHz, 16.8 GHz, and 28 GHz, respectively.

IV. Si-AlN Hybrid System

A. Silicon PICs

Silicon PICs have been intensively investigated for more than two decades. Numerous functional devices have been demonstrated, such as filters [140], [141], polarization beam splitters (PBS) [142], [143], [144], polarization splitters and rotators (PSR) [145], [146], [147], [148], array waveguide grating (AWG) [149], [150], [151], optical switches [20], [21], [22], [23], [24], [26], [48], optical modulators [152], [153], [154], [155], [156], [157], etc. Accompanying the development of devices with different functions, Silicon PICs for different applications appeared, as shown in Fig. 4, including optical interconnect [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [158], [159], optical signal processor [10], [160], [161], [162], [163], [164], light detection and ranging (LiDAR) [165], [166], [167], [168] and optical sensing [1], [2], [3], [4], [5], [6], [7], [8].

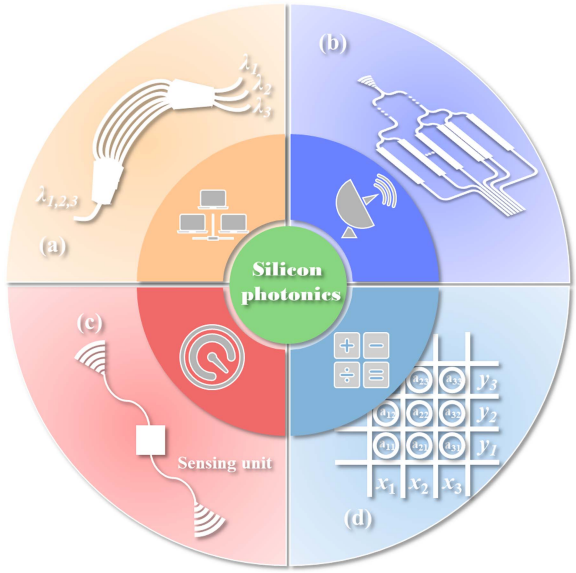


Fig. 4. Applications of SiPh. (a) An AWG as (de)multiplexer in interconnect application. (b) An optical phased array (OPA) for LiDAR systems. (c) An optical sensor with sensing units. (d) A micro-ring resonator array for optical computing.

SiPh was born for high-performance interconnecting purpose, which requires ultrafast modulation speed and high transmission capacity. Currently, a lot of high-speed plasma dispersion effect-based silicon modulators have been demonstrated, showing modulation speeds over 100 Gbps [169], [170]. A recent paper also reported over 240 Gbps [42], [43]. Besides, to increase the transmission capacity, several multiplexing technologies have been demonstrated [15], such as wavelength-division (de)multiplexing (WDM) [18], [171], polarization-division (de)multiplexing (PDM) [16], [17], [172], mode-division (de)multiplexing (MDM) [158]. Dai et al. [15] proposed the concept of multi-dimensional multiplexing technology to improve the link capacity further. proposed the concept of multi-dimensional multiplexing technology. As shown in Fig. 5(a), the system, combining PDM and MDM, offers the potential for ultra-high capacity.

Similar technologies can also be applied to other applications with the development of optical interconnect. By using a high-speed phase shifter, fast-scanning LiDAR systems [165], [166], [167], [168] have been reported. Fig. 5(b) shows an 8×8 optical phased array (OPA) [167]. Each array pixel is controlled by an independent TO phase shifter, enabling OPA to dynamically form arbitrary far-field radiation patterns. Besides, based on the interaction between the evanescent field out of the waveguide core and analytes will affect the light transmission, several different types of sensors, like biochemical sensors, and acoustic sensors, as shown in Fig. 5(c) and (d).

The high-performance (high link capacity and fast modulation speed) interconnect leads to a boom in optical signal processors. Moreover, the high integration of SiPh devices enables ultra-compact massive signal processing systems [173], [174]. Several architectures have been proposed for matrix-vector multiplication (MVM), like Mach-Zender interferometers (MZIs)

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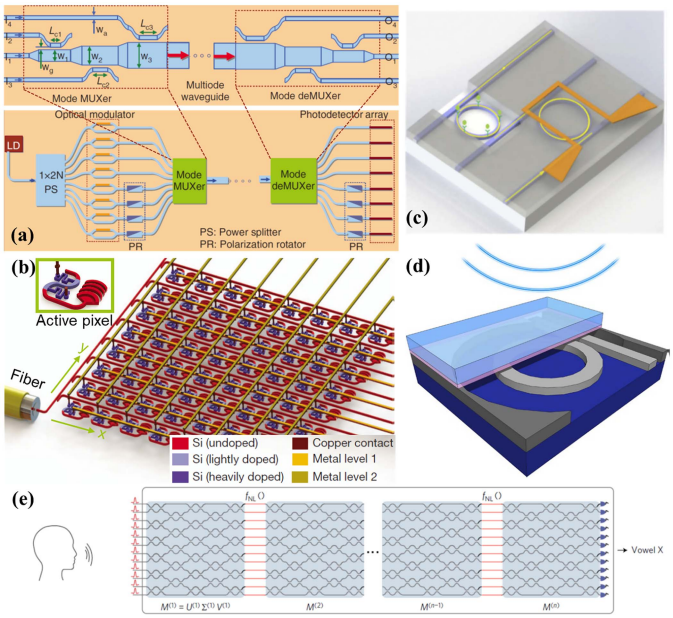


Fig. 5. (a) Schematic of a hybrid multiplexing optical interconnect system using MDM and PDM. Inset: Schematic of MDM [15]. Reproduced with permission. © 2014, De Gruyter. (b) Schematic of optical beam steering and shaping network using an 8×8 optical phased array (OPA). Inset: Schematic of the active pixel [167]. Reproduced with permission. © 2013, Springer Nature. (c) Schematic of a Vernier effect-based tunable mid-infrared sensor [1]. Reproduced with permission. © 2020, The Optical Society. (d) Schematic of an ultrasound sensor [2]. Reproduced with permission. © 2021, Springer Nature. (e) The multilayer structure of an optical neural network (ONN), which can realize vowel recognition [10]. Reproduced with permission. © 2017, Springer Nature.

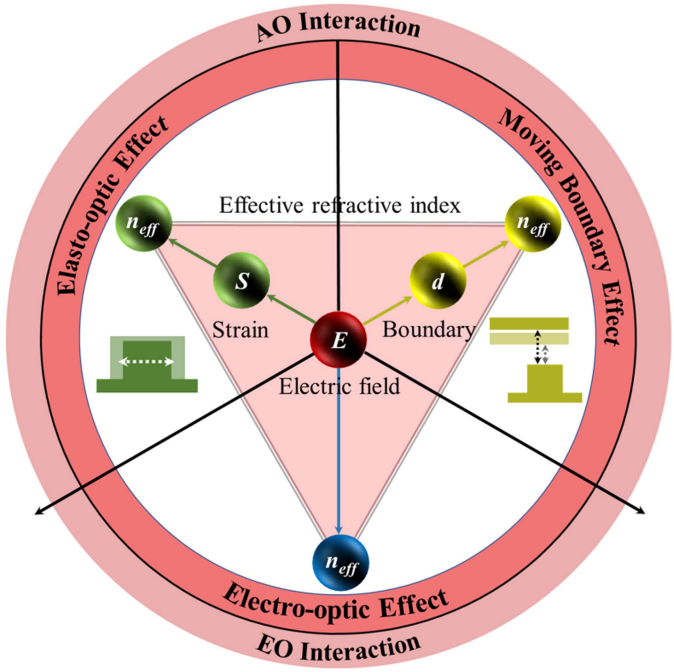


Fig. 6. Interaction mechanisms in Si-AlN hybrid integrated photonics.

array [10], [160], [161], multimode interferometers (MMIs) array [162], and micro-ring resonators (MRRs) array [163], etc. Shen et al. [10] have reported an optical neural network (ONN) consisting of 56 programmable MZIs, shown in Fig. 5(e). Note that, due to the significant power budget of a massive signal processing system and the high loss of the plasma dispersion effect-based devices, TO tuning [175] becomes the primary means for such signal processing systems.

Even though multiple milestones have been achieved on SiPh, as mentioned in the previous section, the silicon material has intrinsic limitations, such as the unavailable Pockels and piezoelectric effect. The hybrid integration concept may overcome such limitations by providing materials with desired functions, thus showing great potential for future applications.

B. Si-AlN Integrated AO Devices

Since wide applications with light and sound interaction have been discovered [176], [177], [178], [179], the mechanism for microwave-optical modulation has attracted significant attention. The devices utilizing AO effect have the advantages of low loss and high modulation efficiency. The strong AO interaction can be achieved in a confined system, especially utilizing acoustic waves with sub-optical wavelengths for ideal mode overlapping. As mentioned in Section II, reactive magnetron sputtering technologies can obtain high-quality AlN films, which provide additional freedom for hybrid integration. A new efficient tuning method, AO interaction, is brought to the SiPh platform. There are two main coupling mechanisms towards the AO interaction, i.e., elasto-optic interaction [87], [89], [180], and moving boundary effect [181], [182], [183]. The moving boundary mechanism can be summarized as the effective index (n_{eff}) of the optical mode changes due to the change in waveguide material refractive index by the moving boundary (the boundary between waveguide, cavity, surrounding media, etc.). The interaction mechanisms are summarized in Fig. 6, which includes elasto-optic, moving boundary, as well as electro-optic effects. Another efficient method for AO interaction is the mechanism utilizing the elasto-optic effect. This mechanism requires the spatial overlap of the acoustic and optical modes, the overlap (g_{ao}) is explained in (1) [87],

$$g_{ao} \propto \frac{\iint \left(n_0^2(\mathrm{r}_\perp)E_2^x(\mathrm{r}_\perp)\right) * E_1^x(\mathrm{r}_\perp)(p_{11}\mathrm{S}_{xx}(\mathrm{r}_\perp)+p_{12}\mathrm{S}_{yy}(\mathrm{r}_\perp))d\mathrm{r}_\perp}{\left(\iint \left|E_2^x(\mathrm{r}_\perp)\right|^2 d\mathrm{r}_\perp \iint \left|E_1^x(\mathrm{r}_\perp)\right|^2 d\mathrm{r}_\perp\right)^{\frac{1}{2}}} \quad (1)$$

where $E_{12}^x(\mathrm{r}_\perp)$ is the electric field along the x-axis of the optical Mode 1 and Mode 2, $\mathrm{S}_{xx,yy}$ is the strain field in x/y direction, p_{11} and p_{12} are the material photoelastic coefficients. The strain (S) field on waveguides modulates the phase of the optical mode by the propagating stress (T) field generated by the piezoelectrically induced acoustic wave.

Conventional AO modulators are based on bulk materials and devices. Since 1970, several works have been reported to achieve AO modulation [184], [185]. However, it faces the challenge of integrating into a highly compact optical system. Due to the strong modulation ability (i.e., high elasto-optic and EO coefficients) and excellent optical properties, $LiNbO_3$ has been reported as an exemplary AO material [186], [187],

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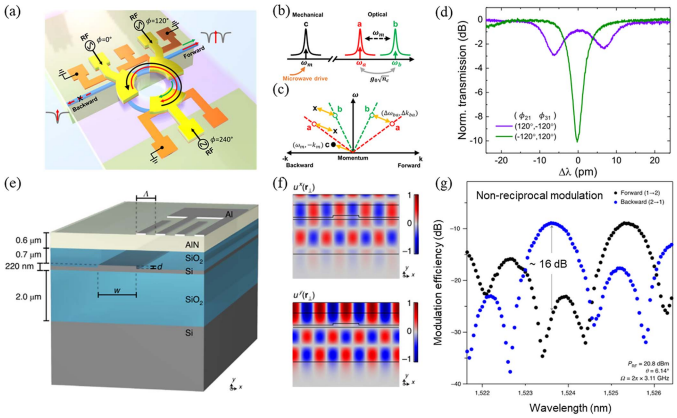


Fig. 7. (a) Schematic of an AlN-based optical isolator. (b) The indirect interband transition in the frequency-domain. (c) The illustration of the phase-matching/-mismatching of optical modes by a rotation acoustic wave in frequency-momentum space. (d) Experimental results of non-reciprocal modulation (a)–(d) Reproduced with permission [90]. © 2021, Springer Nature. (e) Cross-section of an integrated AO modulator with a SiO_2 surrounded silicon rib waveguide and an AlN thin film-based SAW. (f) Simulated displacement profiles (in x and y directions) generated by SAW transducers. (g) Experimental results of the non-reciprocal modulation. (e)–(g) Reproduced with permission [87]. © 2021, Springer Nature.

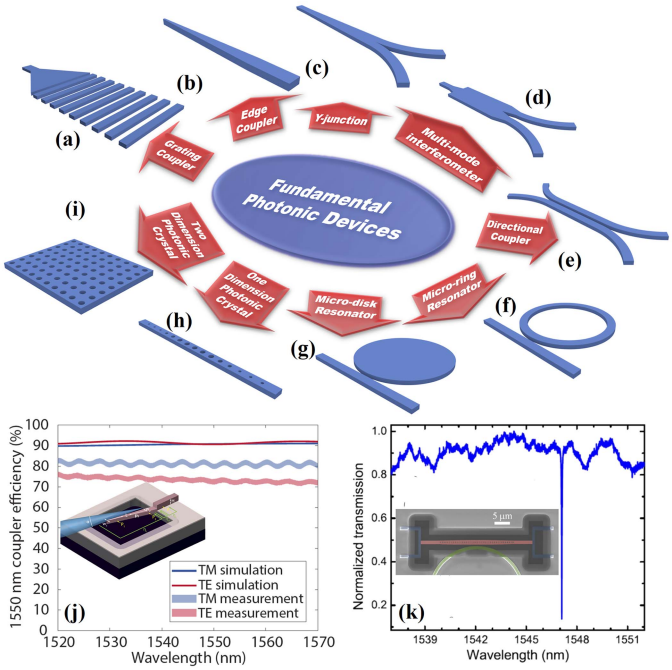


Fig. 8. AlN-based integrated photonic device structures, including (a) grating coupler, (b) edge coupler, (c) y-junction splitter, (d) multi-mode interferometer, (e) directional coupler, (f) micro-ring resonator, (g) micro-disk resonator, (h) one dimension (1D) photonic crystal, (i) two-dimension (2D) photonic crystal. (j) Normalized transmission of suspended inverse taper edge coupler, inset: schematic of an edge coupler. Reproduced with permission [106]. © 2020, The Optical Society. (k) Transmission spectrum of a nanobeam cavity showing the fundamental resonance mode at 1541.7 nm with $Q_{intrinsic}$ of 120000, inset: schematic of a nanobeam cavity. Reproduced with permission [199]. © 2013, AIP Publishing.

[188]. Recently, due to the requirement of on-chip integration with SiPh, AlN thin-film-based piezoelectric transducers for AO modulation have been demonstrated [87], [89], [90], [180]. Because of the structure of the AlN and Si stack, the piezo-induced strain always propagates in the thickness direction. Therefore, a BAW-like transducer becomes one of the most promising candidates in the Si-AlN hybrid system. Meanwhile, some researchers utilize the energy leaked to the substrate from SAW transducers to achieve AO interaction.

The first demonstration of BAW-based AO modulation on SiN_x MRR was reported by Tian et al. in 2020 [89]. Using a 36 dBm of RF power to excite an AlN BAW, a 17 dB isolation contrast between two opposite directions is achieved. Recently, a similar AO modulation device has been reported by the same group [90]. Using a suspended structure, they demonstrated an AO modulation utilizing a high-overtone bulk acoustic resonator (HBAR). As shown in Fig. 7(a), the SiN_x waveguide was placed at a position with maximum mechanical stress. To achieve non-reciprocal transmission, three HBARs operating at equally spaced phases (0°, 120°, 240°) were employed to drive a 2.958 GHz counter clockwise rotational acoustic wave. The phase-matching condition is only satisfied for the counter clockwise transmitting inside the MRR (Fig. 7(b)–(c)), resulting in a maximum of 10 dB isolation (Fig. 7(d)) under 300 mW total RF power.

In 2021, Kittlaus et al. proposed an integrated AO device with silicon waveguides and AlN-based SAW [87], achieving non-reciprocal modulation with centimeter scales interaction length. The schematic of SAW generated by the IDT in the piezoelectric layer propagating to the silicon photonics is illustrated in Fig. 7(e). As shown in Fig. 7(f), the displacement fields generated by the SAW transducers can propagate along the thickness direction, then interact with the optical waveguide. The result of modulation efficiency is shown in Fig. 7(g), exhibiting a non-reciprocal contrast of 16 dB around 1523.7 nm. Different from the work discussed before, non-reciprocal modulation is achieved by tuning the interacting angle between the propagating optical modes and the acoustic wave.

V. AlN Photonic Devices

A. AlN Passive Devices

The AlN photonics platform has several unique advantages compared to the SiPh platform. Due to its large band gap, AlN has a wide transparency window, covering the UV to the mid-IR wavelengths. AlN has a refractive index of ~2.14 [65], a relatively high contrast with SiO_2, making it suitable for building waveguides. An ultralow loss AlN waveguide with ~0.42 dB/cm [94] has been reported. AlN also has a significant second-order optical nonlinear property. The high thermal conductivity and low TO coefficient allow AlN to handle high-power injection well, which is highly beneficial for nonlinear applications [189], [190]. In addition, the piezoelectric effect of AlN allows it to support both acoustic and optical wave propagation in the same layer, significantly improving the AO overlap. As shown in Fig. 8(a)–(i), various basic AlN photonic devices have been reported, such as couplers, resonators, directional couplers, etc.

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Coupler is one of the most fundamental components in PICs. Two coupling approaches, i.e., vertical coupling and edge coupling, have been demonstrated to couple light between PICs and external light sources/detectors. The vertical coupling, shown in Fig. 8(a), uses a pair of grating couplers (GCs) to diffract light from/into optical fibers into/from the PICs. This approach has high alignment tolerance that simplifies testing complexity. In the meantime, it does not require splintering and face grinding and is suitable for wafer-scale testing. Ghosh et al. [191] have reported a 1D fully etched grating coupler with a maximum coupling efficiency (CE) of -6.6 dB for TE polarization and 60 nm 1 dB bandwidth (BW) at the center wavelength of 1550 nm. A focusing GC has been proposed to reduce the device footprint [105]. By precisely controlling the period and duty cycle of a grating, A VIS GC and a NIR GC have been obtained. The TE polarization CE is – (7.7±1) dB at 770 nm with 30 nm bandwidth and – (8.5±1) dB at 1550 nm with 60 nm bandwidth. VIS GC on AlN-on-sapphire [192] was reported in 2018. The peak CE for TE mode is −8.11 dB at 532.9 nm and −8.05 dB at 625.3 nm.

Compared with the GCs, edge couplers (ECs), shown in Fig. 8(b), can provide higher CE, wider BWs, and polarization-independent coupling. The EC uses an inverse taper structure to reduce the large mode mismatch between optical fiber and waveguide. A suspended inverse taper (Fig. 8(j)), optimized for a 600 nm thick AlN layer, has been reported [106]. It achieved high CE, −2.6 dB/−2.6 dB (TE/TM) at 780 nm, −1.55 dB/−0.97 dB (TE/TM) at 1550 nm.

Fig. 8(c)–(e) give typical structures for y junction, multimode interferometer (MMI), and directional coupler (DC), respectively. They are critical components for on-chip beam splitting and routing. A y-junction beam splitter work in the MIR range has been reported [193], showing a uniform 50:50 beam splitting ratio (SR) with less than 2% power difference between two adjacent channels over 200 nm bandwidth (2400-2600 nm). Dong et al. [194] demonstrated a 3 dB MIR MMI with an average SR of – (3.49 ± 0.07) dB from 3.75 μm to 3.85 μm. They also reported a MIR DC at 3.7 μm. The arbitrary SR can be obtained by adjusting the coupling length. To overcome the wavelength and polarization-sensitive issue of conventional DC, a three-waveguide DC has been reported [195]. With two identical gaps between the three waveguides, the SR is always 50:50 because of its symmetric structure. Based on that, by carefully adjusting the intervals, specified SRs, such as 68:32 and 66:34 can be obtained. Besides, by combining two DC with a 50:50 SR, a Mach-Zehnder interferometer (MZI) can be achieved. It has a high extinct ratio (ER), defined as the ratio of the maximum to the minimum at the output, of over 33 dB [195].

Micro-resonators have attracted a lot of interest due to their compact size and strong light confinement. Several micro-resonators, like MRRs (Fig. 8(f)) [54], [59], [65], [86], [91], [104], [105], [107], [108], [109], [192], [194], [196], [197], Micro-disk resonators (MDRs) (Fig. 8(g)) [198], and photonic crystal (PhC) based resonators [199], [200], [201], [202], have been demonstrated.

Among them, high-Q, over 10^5, MRRs [65], [104], [107], [108], [196], [197] operating in different wavelength regimes have been reported. For Q-factor improvement, different deposition technologies, including MOPVE [107], MOCVD [108], etc., were used to improve AlN film quality and reduce propagation loss. Liu et al. [107] reported a high-Q pedestal NIR MRR based on Al-on-sapphire. The intrinsic Q can be up to 2 500 000 for fundamental TE mode with propagation loss of 0.14 dB/cm and 1 900 000 for fundamental TM with propagation loss of 0.2 dB/cm. AlN films were grown by MOVPE, showing excellent crystalline quality. Furthermore, they used multimode microring to minimize field overlap with the waveguide sidewalls and optimize the width of the bus waveguide for phase matching with the fundamental mode. They also reported a high-Q NIR MRR [197] based on the AlN-on-insulator platform, which shows $Q_{\text{intrinsic}} \sim 3\,200\,000$ at 1549 nm for fundamental TE mode and $Q_{\text{intrinsic}} \sim 1\,200\,000$ for fundamental TM mode.

PhC is a periodic dielectric structure (Fig. 8(h) and (i)) that could create an ultracompact resonator. In 2013, Fan et al. [199] reported a suspended 1D-PhC cavity (also called nanobeam). It consists of an elliptical air hole array with a tapering hole size from 700 nm/300 nm to 600 nm/200 nm for the major/minor axis. Based on this structure, the resonance of 1547.1 nm is trapped in the cavity with an intrinsic Q of 120 000 (Fig. 8(k)). Bochmann et al. [200] also reported a suspended nanobeam. The resonance of this nanobeam is at 1531.5 nm with an intrinsic Q of 130 000. In 2015, Li et al. [201] reported another suspended nanobeam with a loaded Q of 51 000. AlN 2D-PhC cavities are rarely explored. Ho et al. [202] demonstrated a 2D-PhC membrane reflector in the MIR regime. With a 2D circle air hole array, the devices can achieve over 90% reflection from 3080 nm to 3780 nm with a peak reflection of 96% at 3160 nm. The TO coefficient extracted from the temperature shift is 2.2×10^{-5} K^{-1}. The relatively low TO coefficient makes AlN good temperature stability, promising for applications with a wide temperature operating range.

We summarize the performance of the AlN-based passive devices in Table IV [65], [104], [105], [106], [107], [108], [189], [191], [192], [193], [196], [197], [198], [199], [200], [201], [202]. With the tuning mechanisms (piezoelectric effect, Pockels effect) that SiPh lacks, AlN shows excellent potential for high-performance active photonic devices, like high-speed, good-linearity, and low-loss optical switches.

B. AlN EO Devices

In contrast to the Si-AlN platform, applying an external electric field on a pure AlN thin film will induce both elasto-optic and EO effects. Therefore, the refractive index change is given [91],

$$\Delta\left(\frac{1}{n^2}\right)_i = p_{ij}\,S_j + r_{ik}E_k \; i,j = 1,2,\ldots,6 \text{ and } k = 1,2,3 \tag{2}$$

Here, p_{ij} is the photoelastic coefficient, S_j is the strain field, r_{ik} is the EO coefficient and E_k is the applied electric field. A proper design, such as avoiding the mechanical resonance frequencies and minimizing strain fields in the device, is required

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TABLE IV
SUMMARY OF ALN PASSIVE DEVICES

Category	Representative Device Performance
Grating coupler [105], [189], [191], [192]	Coupling efficiency: -6.6 dB at λ=1550 nm for TE polarization [191], -4 dB at λ=532.9/ 625.3 nm for TE polarization [192]; 1 dB bandwidth: 60 nm [191]
Edge coupler [106], [192]	Coupling efficiency: <1.99 dB at λ= 360-480 nm for TE polarization [192], -2.6/-2.6 dB at λ=780 nm, -0.97 /-1.55 dB at λ=1550 nm for TM/TE polarizations [106]; Bandwidth: >120 nm [192]
Distributed bragg grating [192]	Extinction ratio: >13 dB at λ=532 nm [192]
Y junction [193]	Propagation loss: 0.83 dB/cm at λ=2400-2600 nm [193]; Bandwidth:>200 nm [193]
Micro-ring resonator [65], [104], [105], [107], [108], [189], [192], [196], [197]	Propagation loss: 0.14/ 0.2 dB/cm for TE/TM polarizations at λ=1550 nm [107], 5.3 dB/cm for TE polarizations at λ=637 nm [192], 8 dB/cm for TE/TM polarizations at λ=390 nm [108]; Q_l [a]: 1400 000/1000 000 for TE/TM polarizations at λ=1550 nm [107]; Q_i [b]: 3200 000/1200 000 for TE/TM polarizations at λ~1550 nm [197]
Micro-disk resonator [198]	Propagation loss: 6.15 dB/cm [198]; Q_i: 56 700, Q_l: 28 350 at λ~1577 nm for TE polarization [198]
Nanobeam cavity [199], [200], [201]	Q_i: 120 000, Q_l: 45 000 at λ~1547.1 nm for TE polarization [199]
2D PhC cavity [202]	Reflection: 96% at λ=3160 nm [202]

a) Q_l: Loaded Q-factor; b) Q_i: Intrinsic Q-factor.

to activate a dominant linear EO property, which is an ideal modulation mechanism we mentioned in the first section.

According to the Pockels effect, the refractive index variation is proportional to the applied electric field, as shown in (3) [93].

$$\Delta n = -\frac{n^3}{2}\sum_{j=1}^{3} r_{ik}E_k \tag{3}$$

The overlap g_{eo} between the electric field and optical field can be extracted from (4). It determines the modulation efficiency [93].

$$g_{eo} \propto \frac{\iint E_{x,op}^2 E_z dxdz}{\iint E_{x,op}^2 dxdz} \tag{4}$$

$E_{x,\,op}$ is the optical field, and E_z is the applied electric field.

Xiong et al. [93] demonstrated an EO modulator consisting of a high-Q (~80 000) MRR and non-planar signal/ground electrodes shown in Fig. 9(a). To reduce the absorption loss from metallic electrodes and have a reasonable EO overlap, they add a 0.8 μm thick oxide layer between the electrodes and the waveguide. Since the highest EO coefficient is along the out-of-plane direction, an electric field along this direction is required. Resonance red-shifted 8 pm when applying DC bias from -7.5 to 7.5 V, as shown in Fig. 9(b). A clear eye diagram (Fig. 9(c)) was observed at the data rate of 4.5 Gbps.

Zhu et al. [59] reported an EO phase shifter with a parallel plate capacitor structure shown in Fig. 9(d). Compared with the non-planar signal/ground structure demonstrated in Ref. [93], it aligns the vertical electric field with the axis of the highest EO coefficient better. However, the measured modulation efficiency is lower than the non-planar signal/ground structure [93]. Two possible reasons may explain this discrepancy: 1) the optical MRRs for the papers are different, which affects the modulation efficiency. 2) The criteria for the electrode designs are different. Ref. [59] set the extra loss of electrodes to 1 dB/cm, while it is ~0.1 dB/cm in [93]. That may result in different distances between electrodes and waveguides and lead to different modulation efficiencies. Zhu et al. further improved the fabrication process [94] and decreased the waveguide propagation loss to 0.42 dB/cm. With the same MRR structure, the $d\lambda_r/dV$ is enhanced from ~0.26 pm/V to ~0.5 pm/V, extracted from Fig. 9(e) and (f). Besides, they reported a push-pull MZI structure, which can reduce the V_π by half, and a dual-MRR structure with ~4 GHz modulation speed.

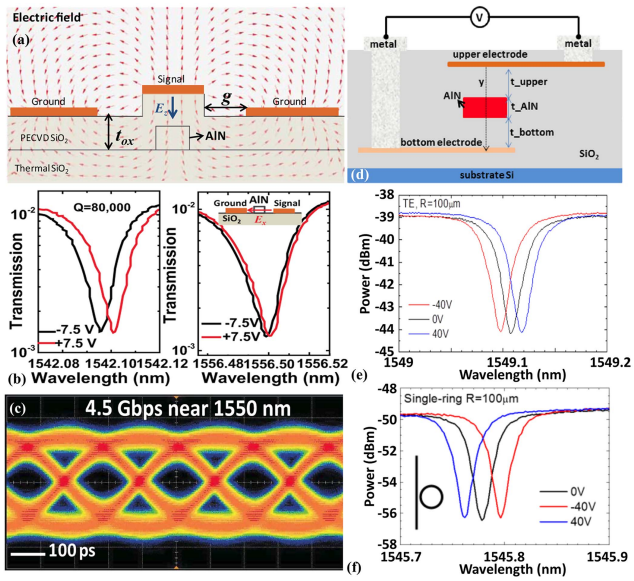


Fig. 9. AlN-based EO modulation. (a) The cross-section and electric field distribution of an AlN modulator with co-planar electrodes. (b) Resonance shifts under different DC biases (from -7.5 V to +7.5 V) when applying vertical (left panel) and horizontal (right panel) electric fields. (c) The eye diagram at 4.5 Gbps near 1550 nm wavelength. The eye extinction ratio is 3 dB with an applied peak to peak voltage of 4 V. (a)–(c) Reproduced with permission [93]. ©2012, American Chemical Society. (d) The cross-section of an AlN modulator with out-of-plane electrodes. (e) Measured spectrum of the MRR under -40, 0 and 40 V bias before fabrication process optimization. (f) Measured spectrum of the MRR under -40, 0 and 40 V bias after fabrication process optimization. (d), (e) Reproduced with permission [59]. © 2016, The Optical Society. (f) Reproduced with permission [94]. © 2019, The Optical Society.

C. AlN AO Devices

As shown in (2), neglecting the EO effect when applying an external electric field requires a minimized overlap between the

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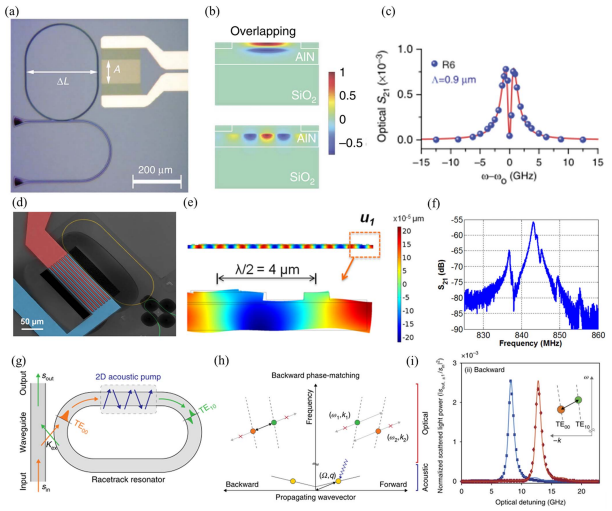


Fig. 10. AlN-based integrated AO modulators using SAW and Lamb wave technologies. (a) Optical image of an AO modulator, including a racetrack resonator and a SAW transducer. (b) Numerically simulated overlap of acoustic mode and optical mode with different IDT pitches. (c) Modulation response of R_6 mode excited by IDTs versus the laser frequency detuning. Here the acoustic wavelength of 0.9 μm. (a)–(c) Reproduced with permission [86]. © 2014, Springer Nature. (d) SEM image of an AO modulator with IDT pitch of 4 μm. (e) The lateral displacement field of the modulator. (f) Measured S_{21} response of the modulator at the optical wavelength of 1616.225 nm with an acoustic signal at the wavelength of 8 μm, the peak value is obtained at 843 MHz. (d)–(f) Reproduced with permission [91]. © 2016, AIP Publishing. (g) A Lamb wave-based non-reciprocal photonic device, consisting of a racetrack resonator supporting TE_{00} and TE_{10} optical modes. (h) Working principle of the mode conversion in the backward directions. (i) The measured non-reciprocal modulation. The phase-matching condition is obtained when the separation of two optical modes equals the applied Lamb wave frequency of 4.55 GHz. (g)–(i) Reproduced with permission [104]. © 2018, Springer Nature.

electric and optical fields. At the pure AlN platform, the acoustic and optical components are on the same layer, so the strain must propagate along with the AlN thin film. Lamb wave thus perfectly matches the requirement with energy confined in the AlN layer. Some research groups have also tried the SAW technology as an alternative approach to achieving AO modulation in the AlN platform [86], [88], [201]. However, drawbacks of using SAW technology include low electromechanical coupling and limited overlap between the acoustic and optical modes.

In 2014, for the first time, the SAW technology was introduced as an approach to producing AO modulation on integrated photonics by Tadesse et al. [86]. The illustration of modulation structure is shown in Fig. 10(a), consisting of an optical racetrack resonator made by AlN rib waveguides and a SAW transducer. The spatial field distribution of acoustic and optical modes is plotted in Fig. 10(b). As a result, when half of an acoustic wave wavelength equals the width of the waveguide, the maximum field overlap can be achieved. Fig. 10(c) shows the AO modulation response of a high-order R_6 mode excited by the SAW versus the laser frequency detuning. The drop at the center frequency indicates a good coupling between the optical resonance and the acoustic wave.

The integration of optical MRR and LWRs has been reported in the last few years [91], [92], [104], [203]. In 2016, an AO modulator based on AlN LWR was first demonstrated by Ghosh et al. [91], using a one-resonator system to achieve reciprocal optical modulation. Fig. 10(d) shows the SEM images of the system, including a racetrack resonator and an LWR. The displacement field of the device is illustrated in Fig. 10(e), in which a rib waveguide is placed at the region with maximum stress. According to their measurement, the mechanical modes in the acoustic domain also appear in the S_{21} optical response, as shown in Fig. 10(f). The optical modulation is characterized, showing a 3.9 pm/V response for the acoustic wave at 843 MHz.

In 2018, a non-reciprocal light transmission technology was developed by Sohn et al. [104], using a Lamb wave transducer. As illustrated in Fig. 10(g), the Lamb wave is applied to the racetrack resonator to break the orthogonality between the TE_{00} and TE_{10} optical modes and achieve the phase-matching condition. For the optical part, the resonance frequency of the TE_{00} mode (ω_{00}) needs to be higher than that of the TE_{10} mode (ω_{10}) in the forward direction or lower in the backward direction to satisfy the phase-matching condition. For the acoustic part, the operating frequency of the generated Lamb wave needs to be the resonance frequency difference between the TE_{00} and TE_{10} modes ($|\omega_{10} - \omega_{00}|$), and the momentum of the Lamb wave should equal the momentum difference between the TE_{00} and TE_{10} modes. The working principle of the non-reciprocal phase-matching is explained in Fig. 10(h). In the backward direction, the anti-Stocks sideband is generated, while the Stocks sideband is greatly surpassed (Fig. 10(i)). Recently, an AO device utilizing a two-resonator system has been demonstrated by Kim et al. [92], with the non-reciprocal optical transmission of ~3 dB contrast ratio.

VI. Summary and Outlook

We have reviewed the recent research progress of integrated AO and EO devices on the AlN platform. Compared with silicon, AlN has an ultra-broad transparency window, covering UV, VIS, NIR, and MIR wavelength regimes. Piezoelectric, nonlinear properties (Pockels effect and second nonlinear effect), and pyroelectric properties are available in this platform due to its centrally asymmetric crystal structure. AlN is an ideal platform for realizing multiphysics couplings, such as AO modulating, EO modulating, and pyroelectric sensing, etc. The fabrication process, including AlN thin film deposition and etching methods, is presented. The fundamental integrated photonic devices (coupler, splitter, resonator, photonic crystal, etc.), acoustic devices (surface acoustic wave resonator, film bulk acoustic resonator, Lamb wave acoustic resonator, etc.), and optomechanical devices (AO tuning device, EO tuning device) are also summarized and discussed. Some typical structures and their experiment results are illustrated in the figures.

In general, we see the following three aspects for future development of AlN-based reconfigurable photonic systems: high-quality AlScN material growth, high-quality AlScN thin-film processing, and potentially reconfigurable PICs with applications covering sensors and quantum systems, as shown in Fig. 11(a).

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TABLE V
COMPARISON OF ALN AND $AL_{1-X}SC_XN$

Properties	AlN	$Al_{1-x}Sc_xN$
$d_{33,p}$ piezoelectric coefficient (pC/N)	~6 [207]	27.6 (x=0.43) [207]
$d_{31,p}$ piezoelectric coefficient (pm/V)	~2.3 [208]	13.5 (x=0.41) [208]
Coefficient of thermal expansion ($\times 10^{-6}$/K)	5.75 at 400 °C [209]	6.38 at 400 °C (x = 0.32) [209]
Thermal conductivity ($Wm^{-1}K^{-1}$)	51 [210]	4.8 (x=0.2) [210]
Elastic modulus (GPa)	535 [209]	270 (x = 0.41) [209]
Effective pyroelectric coefficient ($\mu Cm^{-2}K^{-1}$)	5.40±0.19 [211]	9.98±0.20 (x = 0.3) [211]
Optical waveguide propagation loss (dB/cm)	0.42±0.05 @1550 nm [94]	9±2 @1550 nm (x = 0.15) [212]
$d_{33,o}$ second-order optical susceptibility (pm/V)	5.1±0.4 @1550 nm [213]	62.3±5.6 @1550 nm (x = 0.36) [213]
$d_{31,o}$ second-order optical susceptibility (pm/V)	0.07±0.006 @1550 nm [213]	4.5±0.3 @1550 nm (x = 0.36) [213]
Band gap (eV)	6.1 [95]	4.5 (x = 0.41) [95]
Ferroelectricity	N/A	Yes [a]

a) The coercive electrical field of the $Al_{0.7}Sc_{0.3}N$ thin-film is ~ 3MV/cm [214];

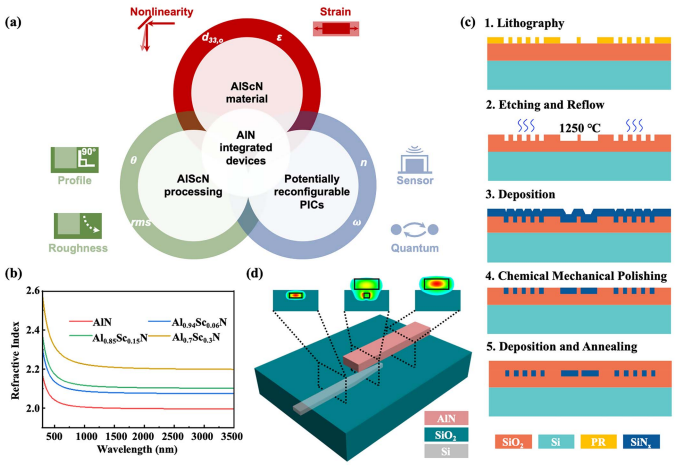


Fig. 11. (a) The schematic of the outlook of AlN integrated devices. (b) The ordinary refractive indices (real part) of AlScN with different compositions. (c) The photonic Damascene process: (1) lithography; (2) etch the substrate to pattern waveguides and reflow with high temperature to smooth the sidewall roughness; (3) core layer deposition; (4) surface polishing; (5) cladding deposition and annealing. (d) Schematic of the vertical coupler between Si and AlN layers with cross-section mode profiles.

A. *Scandium Doped AlN for Properties Enhancement*

As discussed in previous sections, AlN has proved itself as an excellent material for building photonic (MMR, MZI, MMI, etc.), acoustic (SAW resonator, BAW resonator, LWR, etc.), and acoustic-photonic hybrid devices. Especially, it provides a link between the electrical and photonic domains. So far, multiple AlN-based modulation mechanisms have been demonstrated. However, the conversion efficiency from the electrical to photonic domain needs to improve. For example, the AlN-based phase shifter has modulation efficiencies near the 1550 nm wavelength of nearly 240 V·cm for the TE mode and 320 V·cm for the TM mode [59]. Further improving device performance requires a high-quality AlN film with a higher piezoelectric coefficient, pyroelectric coefficient, and optical susceptibility.

Several technologies have been successfully developed to deposit AlScN thin film, i.e., MOCVD, MBE, and magnetron sputtering deposition [100], [204], [205]. ScN is a transition metal nitride semiconductor, which has a rocksalt and nonpolar crystal structure [206]. AlN has a wurtzite structure (polar). The structure of AlScN alloy is transited from wurtzite to metastable hexagonal (≤43 at. % Sc) or nonpolar (>43 at. % Sc). As shown in Table V [94], [95], [207], [208], [209], [210], [211], [212], [213], [214], compared with the AlN, the scandium-doped AlN (AlScN) thin-film possesses three improvements. Firstly, the piezoelectric and pyroelectric properties are greatly improved [208], [211]. Secondly, the second-order optical susceptibility is enhanced by more than one order [213]. Thirdly, the ferroelectric property has been discovered in high concentration doped thin films [214], [215]. Besides, AlScN is also compatible with the CMOS process [216]. For these reasons, AlScN has attracted extensive attention in sensing, signal processing, and RF communication. So far, a variety of AlScN-based piezoelectric MEMS devices have been demonstrated [131], [217], [218]. [219], [220], [221]. However, only a few works investigated the AlScN-based integrated photonic devices [212], [222], showing higher modulation efficiency. In the meantime, much larger optical propagation loss is also observed in AlScN thin films [213], especially for AlScN with high scandium concentration (17.2 dB/cm @1550 nm for $Al_{0.64}Sc_{0.36}N$). The huge loss would severely hinder the development of AlScN photonic devices. Therefore, reducing the propagation loss through improving AlScN film quality and optimizing the etching process is not trivial. We measure the ordinary refractive indices of AlScN with different compositions and plot the real part of them in Fig. 11(b). The refractive index gradually increases for a higher scandium concentration, providing a good choice for waveguiding. The image part of the refractive indices for the four scandium concentrations is below the measurable value, indicating negligible material absorption loss for $Al_{1-x}Sc_xN$. We expect that by solving the scatting loss issue, the AlScN thin film would significantly advance the development of AlN thin-film-based reconfigurable photonic devices.

B. *High-Quality AlN Thin-Film Processing*

Several epitaxial technologies have been reported to achieve strong orientation (FWHM<1°) single crystal AlN film, like MOVPE, MBE, etc. Such AlN film shows good nonlinear/EO/AO properties.

Low loss waveguide is critical in photonic integrated circuits. Etching-induced sidewall roughness generates scattering and significantly deteriorates the propagation loss. Inspired by the "Damascene process" [223] for copper interconnects, Pfeiffer

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et al. [224] demonstrated a high-Q SiN_x MRR using the "photonics Damascene process". As shown in Fig. 11(c), unlike the subtractive process, the waveguide is pre-patterned on the substrate, followed by core material deposition. Then, chemical mechanical polishing (CMP) is used to planarize the surface and remove the excess core material. Finally, the cladding is deposited on the surface, and annealing is used to densify the film. Compared with other SiN_x MRRs fabricated by a typical subtractive process, it shows a high-Q factor of 3.7×10^6 with a compact waveguide of $1.5 \times 1.35\ \mu m^2$. After that, they optimized the process by adding the reflow step [225], shown in Fig. 11(c) step 2. With a temperature of 1250 °C and SiO_2 surface tension, the sidewall roughness caused by etching is reduced, leading to the scattering loss reduction and achieving a mean Q-factor over 5×10^6.

Theoretically, the photonic Damascene process also can be applied to reduce the loss of AlN waveguides, especially for AlScN waveguides. However, growing a high-quality AlN film with a specific orientation on trench corners is challenging. If this tissue could be overcome, the photonic Damascene process will significantly improve the performance of AlN-based photonic devices and bring substantial new applications such as nonlinear photonics, quantum photonics, etc.

C. Potentially Reconfigurable PICs

One key feature of AlN is that it could provide reconfigurability to current PICs. As mentioned in the previous sections, multiple AlN-based reconfigurable devices have been demonstrated, showing great potential in future PIC applications. Among them, there still exists plenty of room for exploring tuning technologies.

MEMS technology could provide large effective index tuning and has been widely investigated in silicon photonics [226], [227], [228], [229], [230]. In 2019, Casset et al. reported a $Pb(Zr_{0.52}Ti_{0.48})O_3$ (PZT) thin-film-based phase shifter on a silicon photonic device [231], which achieves a modulation efficiency ($V_\pi \cdot L$) of 20∼26 V·cm. However, very few paper discussed AlN-based MEMS tuning technology in reconfigurable PICs [181]. The capability to control the interaction between light and motion of micro-/nano- mechanical systems enables fundamental physics to turn from theory to reality. Thanks to the high piezoelectric coefficient, high acoustic velocity, and CMOS process compatibility, AlN-based MEMS technology reveals great potential in quantum technology [232]. We expect more AlN-based MEMS tuning approaches to be developed in the near future.

AlN-based high-speed AO tuning technology has been investigated. However, few papers discuss low-speed tuning with piezoelectric-based flexural devices ($<$ 10 MHz or even electrostatic displacement). This kind of device is suited to the case that does not require high-speed but higher sensitivity and signal-noise ratio (SNR). Recently, Westerveld et al. [2] reported an ultrasound sensor in silicon photonics with extremely high sensitivity using an innovative optomechanical waveguide. The incident ultrasound induces the membrane vibration and changes the gap between the membrane and the waveguide. The gap change results in a variation of the effective index of the guiding mode. AlN-based flexural devices can be applied to strain-based effective index modulation too. Such integration can fully meet the requirement of high sensitivity, high SNR, and small size. Thus, AlN-based flexural resonators for AO modulation deserve further exploration.

AlN thin film is CMOS process compatible and friendly for integration with silicon photonics. Section IV summarizes a variety of reported Si-AlN hybrid devices. Such a hybrid platform is attractive because it combines the advantages of silicon photonics and AlN. Silicon waveguides and the AlN functional layer provide optical guiding and tuning capability, respectively. However, the finite mode-overlap between them limits the tuning efficiency. Such an issue may be solved by 3D hybrid integration, which separates the guiding and modulation layers. As shown in Fig. 11(d), a vertical adiabatic inverse taper could provide a link for these two layers with negligible coupling loss. Modulation signals are loaded on the AlN layer and transmitted in the silicon layer. We expect an improved modulation efficiency for this hybrid Si-AlN system.

Moreover, for pure photonic devices, the large bandgap and broad transparency window of AlN open the possibility for UV light generation [233], [234] and detection [235], [236]. Yoshida et al. [234] have demonstrated an AlGaN multiple-quantum-well (MQW) laser diode (LD), which can lase in 320-340 nm with 3 mW peak power and 1.1% differential external efficiency. By controlling the AlN mole fraction, the lasing wavelength can extend to deep ultraviolet (DUV). Liu et al.[235] reported an ultra-long AlN nanowire photodetector, which can achieve a response of $\sim$ 1 ms, and a photocurrent response of 2.7×10^6 A/W at 325 nm.

Both Si-AlN hybrid and pure AlN systems are excellent platforms for building microwave-mechanical-photonic system. By improving material quality, optimizing processing, and tailoring devices, such piezo-optomechanical devices may hold great promise in quantum technologies [176], [232], [237], [238], [239] since they enable coupling from qubits to a mechanical mode and subsequently convert to the photonic domain that avoids noise from direct conversion from electrical to photonic parts in the superconducting circuits [240].

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